

Charlie Watson

London, United Kingdom · wato95@gmail.com · [Website](#) · [LinkedIn](#) · [GitHub](#)

Data engineer with eight years at the UK financial regulator, building and running the platform its wholesale markets supervision runs on: Python and SQL pipelines orchestrated in Airflow, processing 150M+ transactions a day into a Redshift warehouse of over 600TB. I own it from ingestion and data modelling through validation, performance tuning and the metric definitions the people who act on the data rely on.

PROFESSIONAL EXPERIENCE

Financial Conduct Authority (FCA)

London, United Kingdom

Lead Associate (Data Engineering and Analytics), Market Analysis and Risk

2024-02 – present

- Built and run the end-of-day data platform in Python and SQL, batch processing 150M+ transactions a day across FX, rates, credit, equities and commodities. Orchestrated as 4 Airflow DAGs against a 9am readiness SLA, and migrated from EC2 to SageMaker Unified Studio over a single weekend with a week of dual-running and no missed SLA.
- Wrote over 75 automated validation and outlier rules that catch mis-keyed prices and quantities before they reach anyone who would act on them. Failing rows are quarantined rather than dropped or allowed to block the batch, and surface in a data quality dashboard of open issues that supervisors use to have firms correct their reporting.
- Designed the data model underneath it: a Redshift warehouse of over 600TB layered raw, staging and production across domain tables for prices, curves and surfaces, partitioned schemas and ETL tuned for query and storage cost against an S3 data lake, and over 25 enrichment tables whose distribution and sort keys were chosen for query performance.
- Redesigned the schema of the 300bn-record core table most of the team's analysis runs on, with one colleague: 60 tables rebuilt, 30 taken end to end, with the speed-up coming mainly from query rewrites that exploited the new keys. Daily batch fell from 1.5 hours to 35 minutes, and the cutover ran with no downtime, reconciled on group-level exposure sums per firm rather than row counts alone.
- Ported three dashboard transformations to dbt Core on the production Redshift warehouse, triggered from the existing Airflow DAG with dbt tests on every output. Running in parallel with the original SQL and checked against it daily ahead of cutover.
- Write and maintain the in-house Python package most of the team's pipelines import, standardising Pandas and NumPy utilities, SQL connectors and vendor API integrations across nine people, with contributions arriving as merge requests I review. Automated testing, 47 in pytest covering the database, Bloomberg and Tableau connectors, run by GitLab CI on every merge request.
- Agreed the 12 metrics another team would track before any analysis started, each definition a negotiated compromise between what they wanted and what the data could honestly support, then documented them beside the charts in the dashboard. The team produces every cut themselves now and has come back with no further requests since handover.
- Containerised the four Streamlit applications the department runs for analysts whose problems have outgrown a dashboard, owning the Dockerfiles and the data engineering that feeds them.

Senior Associate (Data Science and Analytics), Market Analysis and Risk

2020-10 – 2024-02

- Asked to review dashboard permissions by hand, found the Tableau REST API instead and built an automated audit in Python across over 150 dashboards, revoking 30–50 stale permissions a month with a one-click path back for anyone who needed access again. Still running years later.
- Told a supervision team asking for transactions with high-risk jurisdictions that the data could go further, then built it unprompted in Python with NetworkX across 100M+ transactions, tracing chains through corporate entities and surfacing three firms they had not identified.

Associate (Analytics and Modelling), Market Analysis and Risk

2018-12 – 2020-10

- Built outlier detection on derivative valuations, screening around 100M daily notional amounts: the method moved from kernel density estimation to local outlier factor to histogram-based outlier score for the throughput required, and fitting thinly-traded subsegments individually rather than applying one global threshold cut false positives by 26% over the following year.
- Spotted that another team was copying data out of free-text emails by hand and replaced it, unasked, with an Excel capture template, a Python extraction routine, a log and a dashboard on the trends in what firms were reporting. They asked for it on two more inboxes, it saves around 15 hours a month, and it is still running six years later.

Research Analyst, Competition and Economics

2018-07 – 2018-12

- Owned the vulnerability workstream of the high-cost credit market study, on a dataset joining two years of transactions for 1.5M current account customers with credit reference files for around 10% of UK adults. The study fed the June 2019 overdraft remedies.

PROJECTS

PulseFoundry and QuakeLens

2026 – present

Batch data-product pipeline over public USGS and US Census data, and the live browser app that consumes it. Sole author. · wato95.github.io/quakelens-web

- Built the pipeline in Python, DuckDB and dbt: raw provider responses archived as the exact bytes received with SHA-256 metadata and idempotent re-capture, append-only event revisions in place of destructive upserts, and source, staging, intermediate and publication dbt models with data-quality tests.
- Publishes versioned Parquet artifacts behind a checksummed manifest, validated and reconciled before an atomic install, which QuakeLens queries directly in the browser with React, TypeScript and DuckDB-Wasm.
- 310 pytest tests, strict Pyright and Ruff, run by GitHub Actions on every pull request, with committed fixtures keeping CI independent of live services.

EDUCATION

University of Leicester

Leicester, United Kingdom

MPhys Physics with Astrophysics

2014-09 – 2018-06

First Class Honours, 85% average over four years. Department scholarship for promising students in Physics. Masters thesis: N-body simulations of galactic mergers.

QUALIFICATIONS

- **CFA Level 1**, CFA Institute. Top 90th percentile, first attempt.

KEY TECHNICAL SKILLS

Python, SQL, dbt, Apache Airflow, AWS (S3, EC2), Amazon Redshift, Docker, Data modelling and schema design, Data quality and validation, pytest, GitLab CI, GitHub Actions, Git and code review, ETL / ELT, Query optimisation, Pandas, DuckDB, Parquet, Tableau, REST APIs, Metric definition